Experiment 6 – Code, Input and Output

1. Module 1 – Prime Number and Factorial

Code: module1.py

```
def check_prime(num):
    if num == 1:
        print(" prime no")
        return
    if num == 2 or num == 3:
        print("prime no")
        return
    if num % 2 == 0 or num % 3 == 0:
        print("Not prime")
        return
    count = 5
    for count in range(5, num, count + 6):
        if (num % count) == 0 or ((num) % count + 2 == 0):
            print("Not prime")
            return
    print("prime no")


def calculate_factorial(num):
    fact = 1
    if num > 0:
        for count in range(1, num + 1):
            fact *= count
        print("\nFactorial of ", num, " is", fact)
    elif num == 0:
        print("\nFactorial of 0 is 1")
    else:
        print("\nPlease enter a non-negative integer !")
```

Code: Main program shown in the PDF

```
from module1 import check_prime

number = int(input("\nEnter number = "))
print(check_prime(number))
print(calculate_factorial(number))
```

Input:

```
Enter number = 5
```

Output shown in the PDF:

```
prime no
None

NameError: name 'calculate_factorial' is not defined
```

Note: The PDF shows this NameError because calculate_factorial is called but is not imported in the displayed main program. The output above is kept as shown in the source PDF.

2. Module 2 – Arithmetic Operations

Code: module2.py

```
def add(a, b):
    result = a + b
    return result

def sub(a, b):
    result = a - b
    return result

def mul(a, b):
    result = a * b
    return result
```

```
def div(a, b):
    result = a / b
    return result
```

Code: Main program shown in the PDF

```
import module2 as m1

a, b = map(int, input("Enter two numbers separated by a space: ").split())
print(m1.add(a, b))
print(m1.mul(a, b))
```

Input:

```
Enter two numbers separated by a space: 2 3
```

Output:

```
5
6
```